

CS 4391 — Group 17

Real-Time Food Detection

on the NVIDIA Jetson Orin Nano

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From tray to data

STARTED HERE

Waste tracking

A diner aims a camera at their returned tray. Over time the kitchen learns that, e.g., peas come back uneaten 3 of 4 times, and orders fewer of them.

Adapt rations → Less food thrown out

FUTURE OBJECTIVE

Calorie estimation

Same front-end, bigger goal: name every dish on the plate → look up calories per portion. Out of scope objectives:

- Needs volume / mass, not 2D boxes
- No depth sensor or scale
- Second model would saturate the Jetson

Both use cases need the same first step, a fine-grained dish detector. That is what we built.

One model, three stages



Model

YOLOv8n — one-stage detector, COCO-pretrained, fine-tuned on our data

Training

100 epochs max, early stop on val mAP (patience 10)

Canteen plates, labelled top-down

Source

Open-source canteen-plate dataset shared with the YOLO food-waste project (Roboflow YOLOv8 export). We use only the labelled images and trained our own model end-to-end.

Plus our own footage

Short clips recorded directly with the Jetson's USB camera, used as a held-out qualitative test set during the live demo.

79

object classes
(plate, cutlery, drinks,
foods, garbage)

~1,500

labelled top-down images
of cafeteria trays

640²

pixel input resolution used
in training and inference

RESULTS — QUANTITATIVE

Average Precision

Results: Full Dataset

0.473

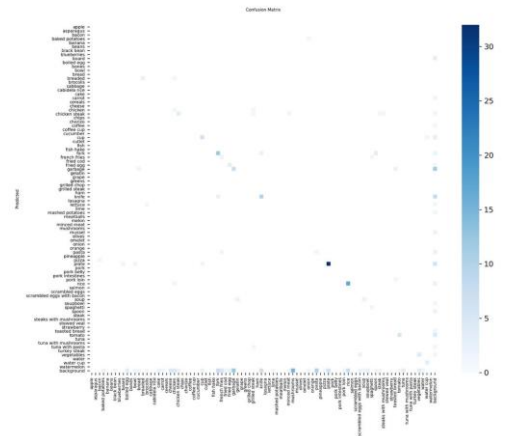
Precision

0.355

Recall

0.333

mAP



Results: Common Variables

0.945

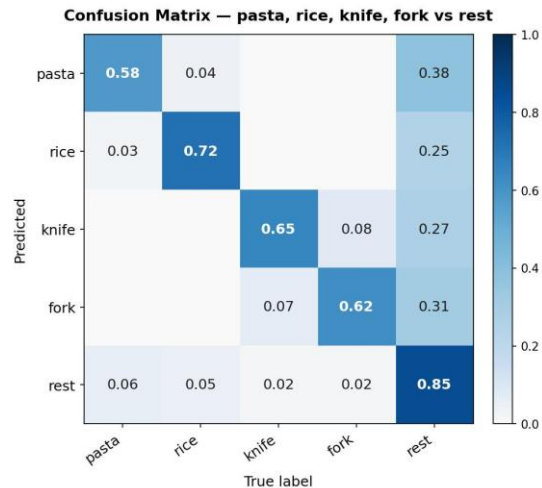
Precision

0.642

Recall

0.716

mAP



Live on the Jetson Orin Nano

TensorRT FP16

best.pt exported on the Jetson itself to a hardware-tuned engine — same Ultralytics interface, faster and steadier inference.

Live USB camera

GStreamer / V4L2 capture, on-screen boxes, FPS overlay, optional --record and --no-display for headless SSH demos.

Next: calories

Detector is the front-end. Calorie head needs depth/scale + per-class density table — future work, beyond Jetson's compute envelope.